

Figure 6: Text size options

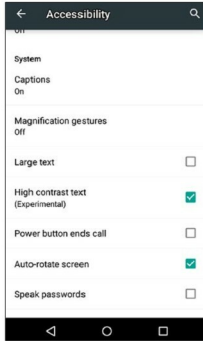


Figure 7: High contrast text

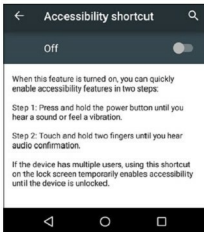


Figure 9: Accessibility shortcut options

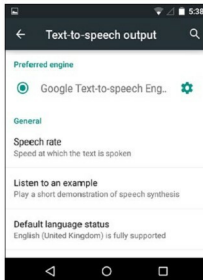


Figure 10: Text-to-speech output

will be increased. This improves the perceivable aspect of the app (Figure 6).

6. **High contrast text:** This option highlights the text so that it becomes easy for people with visual impairment to read it on the screen. Here, light text colours are converted to their darker shades as can be seen in Figures 6 and 7. The text (Experimental) that was grey in Figure 6 is black in Figure 7 after selecting the 'High contrast text' option. Combined with the 'Large text' option, the app makes it easier

to perceive text (Figure 7).

7. **Power button ends call:** As the name indicates, the user can press the power button to end the call, instead of determining the position of the end call button on the touch screen, making the device more operable.
8. **Auto-rotate screen:** This option automatically changes the orientation of the app depending on the orientation of the device. However, if an app has been developed in



Figure 8: Auto-rotate screen

such a way that it runs only in one mode (landscape or portrait), this option cannot change its orientation. Auto-rotate makes the contents on the screen more perceivable (Figure 8).

9. **Speak passwords:** This option reads aloud the password entered by the user, making the app more operable.
10. **Accessibility short cut:** If this option is selected, the user can enable accessibility features using short cuts such as pressing the power button for a few seconds and after hearing a sound, touching the screen with two fingers and holding it till an audio feedback is received. This improves the operability of the device (Figure 9).
11. **Text-to-speech output:** This option enables users to select the text-to-speech engine of their choice. Google is the default engine. The user can select the language and the pace at which the text should be read out. This helps improve the perceivability and understandable aspects of the screen's contents (Figure 10).
12. **Touch and hold delay:** This enables the user to select the duration for which an option should be touched and held in order to activate it. This can be used by people with motor impairment, and helps improve the operability of the apps installed on the device (Figure 11).
13. **Colour inversion:** As the name indicates, the colours on the screen are converted to their respective inverse colours, thus leading to a more perceivable app. This option is also available in the notification bar.
14. **Colour correction:** This option allows users to change the mode of colours on the screen according